

Chapter 3

Numerical Methods

IN THIS CHAPTER we study numerical methods for solving a first order differential equation

$$y' = f(x, y).$$

SECTION 3.1 deals with *Euler's method*, which is really too crude to be of much use in practical applications. However, its simplicity allows for an introduction to the ideas required to understand the better methods discussed in the other two sections.

SECTION 3.2 discusses improvements on Euler's method.

SECTION 3.3 deals with the *Runge-Kutta* method, perhaps the most widely used method for numerical solution of differential equations.

3.1 Euler's Method

If an initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0 \quad (3.1.1)$$

can't be solved analytically, it's necessary to resort to numerical methods to obtain useful approximations to a solution of (3.1.1). We'll consider such methods in this chapter.

We're interested in computing approximate values of the solution of (3.1.1) at equally spaced points $x_0, x_1, \dots, x_n = b$ in an interval $[x_0, b]$. Thus,

$$x_i = x_0 + ih, \quad i = 0, 1, \dots, n,$$

where

$$h = \frac{b - x_0}{n}.$$

We'll denote the approximate values of the solution at these points by $y_0, y_1, \dots, y_n$; thus, y_i is an approximation to $y(x_i)$. We'll call

$$e_i = y(x_i) - y_i$$

the *error at the i th step*. Because of the initial condition $y(x_0) = y_0$, we'll always have $e_0 = 0$. However, in general $e_i \neq 0$ if $i > 0$.

We encounter two sources of error in applying a numerical method to solve an initial value problem:

- The formulas defining the method are based on some sort of approximation. Errors due to the inaccuracy of the approximation are called *truncation errors*.
- Computers do arithmetic with a fixed number of digits, and therefore make errors in evaluating the formulas defining the numerical methods. Errors due to the computer's inability to do exact arithmetic are called *roundoff errors*.

Since a careful analysis of roundoff error is beyond the scope of this book, we'll consider only truncation errors.

Euler's Method

The simplest numerical method for solving (3.1.1) is *Euler's method*. This method is so crude that it is seldom used in practice; however, its simplicity makes it useful for illustrative purposes.

Euler's method is based on the assumption that the tangent line to the integral curve of (3.1.1) at $(x_i, y(x_i))$ approximates the integral curve over the interval $[x_i, x_{i+1}]$. Since the slope of the integral curve of (3.1.1) at $(x_i, y(x_i))$ is $y'(x_i) = f(x_i, y(x_i))$, the equation of the tangent line to the integral curve at $(x_i, y(x_i))$ is

$$y = y(x_i) + f(x_i, y(x_i))(x - x_i). \quad (3.1.2)$$

Setting $x = x_{i+1} = x_i + h$ in (3.1.2) yields

$$y_{i+1} = y(x_i) + hf(x_i, y(x_i)) \quad (3.1.3)$$

as an approximation to $y(x_{i+1})$. Since $y(x_0) = y_0$ is known, we can use (3.1.3) with $i = 0$ to compute

$$y_1 = y_0 + hf(x_0, y_0).$$

However, setting $i = 1$ in (3.1.3) yields

$$y_2 = y(x_1) + hf(x_1, y(x_1)),$$

which isn't useful, since we *don't know* $y(x_1)$. Therefore we replace $y(x_1)$ by its approximate value y_1 and redefine

$$y_2 = y_1 + hf(x_1, y_1).$$

Having computed y_2 , we can compute

$$y_3 = y_2 + hf(x_2, y_2).$$

In general, Euler's method starts with the known value $y(x_0) = y_0$ and computes $y_1, y_2, \dots, y_n$ successively by with the formula

$$y_{i+1} = y_i + hf(x_i, y_i), \quad 0 \leq i \leq n-1. \quad (3.1.4)$$

The next example illustrates the computational procedure indicated in Euler's method.

EXAMPLE 3.1.1. Use Euler's method with $h = 0.1$ to find approximate values for the solution of the initial value problem

$$y' + 2y = x^3 e^{-2x}, \quad y(0) = 1 \quad (3.1.5)$$

at $x = 0.1, 0.2, 0.3$.

SOLUTION We rewrite (3.1.5) as

$$y' = -2y + x^3 e^{-2x}, \quad y(0) = 1,$$

which is of the form (3.1.1), with

$$f(x, y) = -2y + x^3 e^{-2x}, \quad x_0 = 0, \text{ and } y_0 = 1.$$

Euler's method yields

$$\begin{aligned} y_1 &= y_0 + hf(x_0, y_0) \\ &= 1 + (.1)f(0, 1) = 1 + (.1)(-2) = .8, \\ y_2 &= y_1 + hf(x_1, y_1) \\ &= .8 + (.1)f(.1, .8) = .8 + (.1)(-2(.8) + (.1)^3 e^{-.2}) = .640081873, \\ y_3 &= y_2 + hf(x_2, y_2) \\ &= .640081873 + (.1)(-2(.640081873) + (.2)^3 e^{-.4}) = .512601754. \blacksquare \end{aligned}$$

We've written the details of these computations to ensure that you understand the procedure. However, in the rest of the examples as well as the exercises in this chapter, we'll assume that you can use a programmable calculator or a computer to carry out the necessary computations.

Table 3.1.1: Numerical solution of $y' + 2y = x^3 e^{-2x}$, $y(0) = 1$, by Euler's method.

h	x				
	0.2	0.4	0.6	0.8	1.0
0.1	0.640 081 873	0.411 563 195	0.270 299 502	0.186 654 593	0.139 778 910
0.05	0.656 266 437	0.432 887 056	0.291 404 256	0.205 105 754	0.154 715 925
0.025	0.663 635 953	0.442 774 766	0.301 359 885	0.213 956 311	0.162 003 293
Exact	0.670 588 174	0.452 204 669	0.310 952 904	0.222 570 721	0.169 169 104

Examples Illustrating The Error in Euler's Method

EXAMPLE 3.1.2. Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 2y = x^3 e^{-2x}, \quad y(0) = 1$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution

$$y = \frac{e^{-2x}}{4}(x^4 + 4), \quad (3.1.6)$$

which can be obtained by the method of Section 2.1. (Verify.)

SOLUTION Table 3.1.1 shows the values of the exact solution (3.1.6) at the specified points, and the approximate values of the solution at these points obtained by Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$. In examining this table, keep in mind that the approximate values in the row corresponding to $h = .05$ are actually the results of 20 steps with Euler's method. We haven't listed the estimates of the solution obtained for $x = 0.05, 0.15, \dots$, since there's nothing to compare them with in the row corresponding to $h = 0.1$. Similarly, the approximate values in the row corresponding to $h = 0.025$ are actually the results of 40 steps with Euler's method.

You can see from Table 3.1.1 that decreasing the step size improves the accuracy of Euler's method. For example,

$$y_{\text{exact}}(1) - y_{\text{approx}}(1) \approx \begin{cases} .0293 & \text{with } h = 0.1, \\ .0144 & \text{with } h = 0.05, \\ .0071 & \text{with } h = 0.025. \end{cases}$$

Based on this scanty evidence, you might guess that the error in approximating the exact solution at a *fixed value of* x by Euler's method is roughly halved when the step size is halved. You can find more evidence to support this conjecture by examining Table 3.1.2, which lists the approximate values of $y_{\text{exact}} - y_{\text{approx}}$ at $x = 0.1, 0.2, \dots, 1.0$.

EXAMPLE 3.1.3. Tables 3.1.3 and 3.1.4 show analogous results for the nonlinear initial value problem

$$y' = -2y^2 + xy + x^2, \quad y(0) = 1, \quad (3.1.7)$$

except in this case we can't solve (3.1.7) exactly. The results in the "Exact" row were obtained by using a more accurate numerical method known as the *Runge-Kutta* method with a small step size. ■

Table 3.1.2: Errors in approximate solutions of $y' + 2y = x^3e^{-2x}$, $y(0) = 1$, obtained by Euler's method.

h	x									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	0.0187	0.0305	0.0373	0.0406	0.0415	0.0406	0.0386	0.0359	0.0327	0.0293
0.05	0.0087	0.0143	0.0176	0.0193	0.0198	0.0195	0.0186	0.0174	0.0160	0.0144
0.025	0.0042	0.0069	0.0085	0.0094	0.0097	0.0095	0.0091	0.0086	0.0079	0.0071

Table 3.1.3: Numerical solution of $y' = -2y^2 + xy + x^2$, $y(0) = 1$, by Euler's method.

h	x				
	0.2	0.4	0.6	0.8	1.0
0.1	0.681 000 000	0.559 628 676	0.529 820 120	0.569 732 776	0.675 192 037
0.05	0.707 795 377	0.587 454 526	0.557 143 535	0.596 951 988	0.701 764 495
0.025	0.719 226 253	0.600 045 701	0.569 824 171	0.609 684 903	0.714 151 626
"Exact"	0.729 641 890	0.611 901 791	0.581 942 225	0.621 907 458	0.726 015 790

Since we think it's important in evaluating the accuracy of the numerical methods that we'll be studying in this chapter, we often include a row listing values of the exact solution of the initial value problem, even if the directions in the example or exercise don't specifically call for it. If quotation marks are included in the heading, the values were obtained by applying the Runge-Kutta method in a way that's explained in Section 3.3. If quotation marks are not included, the values were obtained from a known formula for the solution. In either case, the shown values are correct.

Truncation Error in Euler's Method

Consistent with the results indicated in Tables 3.1.1 to 3.1.4, we'll now show that under reasonable assumptions on f there's a constant K such that the error in approximating the solution of the

Table 3.1.4: Errors in approximate solutions of $y' = -2y^2 + xy + x^2$, $y(0) = 1$, obtained by Euler's method.

h	x									
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.1	0.0376	0.0486	0.0517	0.0523	0.0522	0.0521	0.0522	0.0522	0.0519	0.0508
0.05	0.0162	0.0218	0.0238	0.0244	0.0247	0.0248	0.0249	0.0250	0.0248	0.0243
0.025	0.0076	0.0104	0.0115	0.0119	0.0121	0.0121	0.0122	0.0122	0.0121	0.0119

initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0,$$

at a given point $b > x_0$ by Euler's method with step size $h = (b - x_0)/n$ satisfies the inequality

$$|y(b) - y_n| \leq Kh,$$

where K is a constant independent of n .

There are two sources of error (not counting roundoff) in Euler's method:

1. The error committed in approximating the integral curve by the tangent line (3.1.2) over the interval $[x_i, x_{i+1}]$.
2. The error committed in replacing $y(x_i)$ by y_i in (3.1.2) and using (3.1.4) rather than (3.1.2) to compute y_{i+1} .

Euler's method assumes that y_{i+1} defined in (3.1.2) is an approximation to $y(x_{i+1})$. We call the error in this approximation the *local truncation error at the i th step*, and denote it by T_i ; thus,

$$T_i = y(x_{i+1}) - y(x_i) - hf(x_i, y(x_i)). \quad (3.1.8)$$

We'll now use *Taylor's theorem* to estimate T_i , assuming for simplicity that f , f_x , and f_y are continuous and bounded for all (x, y) . Then y'' exists and is bounded on $[x_0, b]$. To see this, we differentiate

$$y'(x) = f(x, y(x))$$

to obtain

$$\begin{aligned} y''(x) &= f_x(x, y(x)) + f_y(x, y(x))y'(x) \\ &= f_x(x, y(x)) + f_y(x, y(x))f(x, y(x)). \end{aligned}$$

Since we assumed that f , f_x and f_y are bounded, there's a constant M such that

$$|f_x(x, y(x)) + f_y(x, y(x))f(x, y(x))| \leq M, \quad x_0 < x < b,$$

which implies that

$$|y''(x)| \leq M, \quad x_0 < x < b. \quad (3.1.9)$$

Since $x_{i+1} = x_i + h$, Taylor's theorem implies that

$$y(x_{i+1}) = y(x_i) + hy'(x_i) + \frac{h^2}{2}y''(\tilde{x}_i),$$

where $\tilde{x}_i$ is some number between x_i and x_{i+1} . Since $y'(x_i) = f(x_i, y(x_i))$ this can be written as

$$y(x_{i+1}) = y(x_i) + hf(x_i, y(x_i)) + \frac{h^2}{2}y''(\tilde{x}_i),$$

or, equivalently,

$$y(x_{i+1}) - y(x_i) - hf(x_i, y(x_i)) = \frac{h^2}{2}y''(\tilde{x}_i).$$

Comparing this with (3.1.8) shows that

$$T_i = \frac{h^2}{2}y''(\tilde{x}_i).$$

Recalling (3.1.9), we can establish the bound

$$|T_i| \leq \frac{Mh^2}{2}, \quad 1 \leq i \leq n. \quad (3.1.10)$$

Although it may be difficult to determine the constant M , what is important is that there's an M such that (3.1.10) holds. We say that the local truncation error of Euler's method is *of order* h^2 , which we write as $O(h^2)$.

Note that the magnitude of the local truncation error in Euler's method is determined by the second derivative y'' of the solution of the initial value problem. Therefore the local truncation error will be larger where $|y''|$ is large, or smaller where $|y''|$ is small.

Since the local truncation error for Euler's method is $O(h^2)$, it's reasonable to expect that halving h reduces the local truncation error by a factor of 4. This is true, but halving the step size also requires twice as many steps to approximate the solution at a given point. To analyze the overall effect of truncation error in Euler's method, it's useful to derive an equation relating the errors

$$e_{i+1} = y(x_{i+1}) - y_{i+1} \quad \text{and} \quad e_i = y(x_i) - y_i.$$

To this end, recall that

$$y(x_{i+1}) = y(x_i) + hf(x_i, y(x_i)) + T_i \quad (3.1.11)$$

and

$$y_{i+1} = y_i + hf(x_i, y_i). \quad (3.1.12)$$

Subtracting (3.1.12) from (3.1.11) yields

$$e_{i+1} = e_i + h[f(x_i, y(x_i)) - f(x_i, y_i)] + T_i. \quad (3.1.13)$$

The last term on the right is the local truncation error at the i th step. The other terms reflect the way errors made at *previous steps* affect e_{i+1} . Since $|T_i| \leq Mh^2/2$, we see from (3.1.13) that

$$|e_{i+1}| \leq |e_i| + h|f(x_i, y(x_i)) - f(x_i, y_i)| + \frac{Mh^2}{2}. \quad (3.1.14)$$

Since we assumed that f_y is continuous and bounded, the mean value theorem implies that

$$f(x_i, y(x_i)) - f(x_i, y_i) = f_y(x_i, y_i^*)(y(x_i) - y_i) = f_y(x_i, y_i^*)e_i,$$

where y_i^* is between y_i and $y(x_i)$. Therefore

$$|f(x_i, y(x_i)) - f(x_i, y_i)| \leq R|e_i|$$

for some constant R . From this and (3.1.14),

$$|e_{i+1}| \leq (1 + Rh)|e_i| + \frac{Mh^2}{2}, \quad 0 \leq i \leq n-1. \quad (3.1.15)$$

For convenience, let $C = 1 + Rh$. Since $e_0 = y(x_0) - y_0 = 0$, applying (3.1.15) repeatedly yields

$$\begin{aligned} |e_1| &\leq \frac{Mh^2}{2} \\ |e_2| &\leq C|e_1| + \frac{Mh^2}{2} \leq (1 + C)\frac{Mh^2}{2} \\ |e_3| &\leq C|e_2| + \frac{Mh^2}{2} \leq (1 + C + C^2)\frac{Mh^2}{2} \\ &\vdots \\ |e_n| &\leq C|e_{n-1}| + \frac{Mh^2}{2} \leq (1 + C + \dots + C^{n-1})\frac{Mh^2}{2}. \end{aligned} \quad (3.1.16)$$

Recalling the formula for the sum of a geometric series, we see that

$$1 + C + \dots + C^{n-1} = \frac{1 - C^n}{1 - C} = \frac{(1 + Rh)^n - 1}{Rh}$$

(since $C = 1 + Rh$). From this and (3.1.16),

$$|y(b) - y_n| = |e_n| \leq \frac{(1 + Rh)^n - 1}{R} \frac{Mh}{2}. \quad (3.1.17)$$

Since Taylor's theorem implies that

$$1 + Rh < e^{Rh}$$

(verify),

$$(1 + Rh)^n < e^{nRh} = e^{R(b-x_0)} \quad (\text{since } nh = b - x_0).$$

This and (3.1.17) imply that

$$|y(b) - y_n| \leq Kh, \quad (3.1.18)$$

with

$$K = M \frac{e^{R(b-x_0)} - 1}{2R}.$$

Because of (3.1.18) we say that the **global truncation error of Euler's method is of order h** , which we write as $O(h)$.

Semilinear Equations and Variation of Parameters

An equation that can be written in the form

$$y' + p(x)y = h(x, y) \quad (3.1.19)$$

with $p \neq 0$ is said to be **semilinear**. (Of course, (3.1.19) is linear if h is independent of y .) One way to apply Euler's method to an initial value problem

$$y' + p(x)y = h(x, y), \quad y(x_0) = y_0 \quad (3.1.20)$$

for (3.1.19) is to think of it as

$$y' = f(x, y), \quad y(x_0) = y_0,$$

where

$$f(x, y) = -p(x)y + h(x, y).$$

However, we can also start by applying variation of parameters to (3.1.20), as in Sections 2.1 and 2.4; thus, we write the solution of (3.1.20) as $y = uy_1$, where y_1 is a nontrivial solution of the complementary equation $y' + p(x)y = 0$. Then $y = uy_1$ is a solution of (3.1.20) if and only if u is a solution of the initial value problem

$$u' = h(x, uy_1(x))/y_1(x), \quad u(x_0) = y(x_0)/y_1(x_0). \quad (3.1.21)$$

We can apply Euler's method to obtain approximate values $u_0, u_1, \dots, u_n$ of this initial value problem, and then take

$$y_i = u_i y_1(x_i)$$

as approximate values of the solution of (3.1.20). We'll call this procedure the **Euler semilinear method**.

The next two examples show that the Euler and Euler semilinear methods may yield drastically different results.

Table 3.1.5: Numerical solution of $y' - 2xy = 1$, $y(0) = 3$, with Euler's method.

h	x				
	0.4	0.8	1.2	1.6	2.0
0.2	3.656 000 000	5.706 790 400	11.026 148 66	25.969 172 02	73.797 840 4
0.1	3.802 028 800	6.249 191 282	13.064 051 39	34.570 423 76	115.440 048 3
0.05	3.881 421 103	6.570 796 235	14.405 772 07	41.033 441 26	152.363 866 6
"Exact"	3.966 059 348	6.936 700 945	16.067 111 68	50.000 377 78	211.954 462 2

EXAMPLE 3.1.4. In [Example 2.1.7](#) we had to leave the solution of the initial value problem

$$y' - 2xy = 1, \quad y(0) = 3 \quad (3.1.22)$$

in the form

$$y = e^{x^2} \left(3 + \int_0^x e^{-t^2} dt \right) \quad (3.1.23)$$

because it was impossible to evaluate this integral exactly in terms of elementary functions. Use step sizes $h = 0.2$, $h = 0.1$, and $h = 0.05$ to find approximate values of the solution of [\(3.1.22\)](#) at $x = 0, 0.2, 0.4, 0.6, \dots, 2.0$ by (a) Euler's method; (b) the Euler semilinear method.

SOLUTION(a) Rewriting [\(3.1.22\)](#) as

$$y' = 1 + 2xy, \quad y(0) = 3 \quad (3.1.24)$$

and applying Euler's method with $f(x, y) = 1 + 2xy$ yields the results shown in [Table 3.1.5](#). Because of the large differences between the estimates obtained for the three values of h , it would be clear that these results are useless even if the "exact" values were not included in the table.

It's easy to see why Euler's method yields such poor results. Recall that the constant M in [\(3.1.10\)](#) — which plays an important role in determining the local truncation error in Euler's method — must be an upper bound for the values of the second derivative y'' of the solution of the initial value problem [\(3.1.22\)](#) on $(0, 2)$. The problem is that y'' assumes very large values on this interval. To see this, we differentiate [\(3.1.24\)](#) to obtain

$$y''(x) = 2y(x) + 2xy'(x) = 2y(x) + 2x(1 + 2xy(x)) = 2(1 + 2x^2)y(x) + 2x,$$

where the second equality follows again from [\(3.1.24\)](#). Since [\(3.1.23\)](#) implies that $y(x) > 3e^{x^2}$ if $x > 0$,

$$y''(x) > 6(1 + 2x^2)e^{x^2} + 2x, \quad x > 0.$$

For example, letting $x = 2$ shows that $y''(2) > 2952$.

SOLUTION(b) Since $y_1 = e^{x^2}$ is a solution of the complementary equation $y' - 2xy = 0$, we can apply the Euler semilinear method to [\(3.1.22\)](#), with

$$y = ue^{x^2} \quad \text{and} \quad u' = e^{-x^2}, \quad u(0) = 3.$$

The results listed in [Table 3.1.6](#) are clearly better than those obtained by Euler's method.

Table 3.1.6: Numerical solution of $y' - 2xy = 1$, $y(0) = 3$, by the Euler semilinear method.

h	x				
	0.4	0.8	1.2	1.6	2.0
0.2	3.980 734 157	7.021 003 417	16.381 180 09	51.183 323 26	217.301 032 8
0.1	3.974 067 628	6.980 190 891	16.226 146 39	50.594 503 86	214.631 041 9
0.05	3.970 230 415	6.958 779 586	16.147 129 07	50.298 106 66	213.293 583 0
“Exact”	3.966 059 348	6.936 700 945	16.067 111 68	50.000 377 78	211.954 462 2

Table 3.1.7: Numerical solution of $y' - 2y = x/(1 + y^2)$, $y(1) = 7$, by Euler’s method.

h	x				
	1.2	1.4	1.6	1.8	2.0
0.1	10.083 936 45	14.523 152 45	20.914 624 47	30.117 766 63	43.369 967 16
0.05	10.252 570 17	15.012 952 42	21.981 638 49	32.183 941 34	47.120 835 25
0.025	10.346 014 10	15.287 872 10	22.588 266 22	33.373 738 94	49.308 511 13
“Exact”	10.446 546 23	15.586 440 43	23.253 292 36	34.690 375 09	51.752 229 66

We can’t give a general procedure for determining in advance whether Euler’s method or the semilinear Euler method will produce better results for a given semilinear initial value problem (3.1.19). As a rule of thumb, the Euler semilinear method will yield better results than Euler’s method if $|u''|$ is small on $[x_0, b]$, while Euler’s method yields better results if $|u''|$ is large on $[x_0, b]$. In many cases the results obtained by the two methods don’t differ appreciably. However, we propose the an intuitive way to decide which is the better method: Try both methods with multiple step sizes, as we did in Example 3.1.4, and accept the results obtained by the method for which the approximations change less as the step size decreases.

EXAMPLE 3.1.5. Applying Euler’s method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to the initial value problem

$$y' - 2y = \frac{x}{1 + y^2}, \quad y(1) = 7 \quad (3.1.25)$$

on $[1, 2]$ yields the results in Table 3.1.7. Applying the Euler semilinear method with

$$y = ue^{2x} \quad \text{and} \quad u' = \frac{xe^{-2x}}{1 + u^2e^{4x}}, \quad u(1) = 7e^{-2}$$

yields the results in Table 3.1.8. Since the latter are clearly less dependent on step size than the former, we conclude that the Euler semilinear method is better than Euler’s method for (3.1.25). This conclusion is supported by comparing the approximate results obtained by the two methods with the “exact” values of the solution.

EXAMPLE 3.1.6. Applying Euler’s method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to the initial value problem

$$y' + 3x^2y = 1 + y^2, \quad y(2) = 2 \quad (3.1.26)$$

Table 3.1.8: Numerical solution of $y' - 2y = x/(1 + y^2)$, $y(1) = 7$, by the Euler semilinear method.

h	x				
	1.2	1.4	1.6	1.8	2.0
0.1	10.447 568 67	15.588 535 14	23.256 721 64	34.695 649 22	51.760 178 45
0.05	10.447 038 55	15.587 448 60	23.254 942 52	34.692 912 77	51.756 054 13
0.025	10.446 787 65	15.586 934 68	23.254 101 25	34.691 618 98	51.754 104 26
“Exact”	10.446 546 23	15.586 440 43	23.253 292 36	34.690 375 09	51.752 229 66

Table 3.1.9: Numerical solution of $y' + 3x^2y = 1 + y^2$, $y(2) = 2$, by Euler's method.

h	x				
	2.2	2.4	2.6	2.8	3.0
0.1	0.068 700 000	0.059 732 621	0.050 560 917	0.042 925 892	0.035 985 548
0.05	0.122 879 586	0.061 338 956	0.051 465 256	0.043 967 002	0.038 044 692
0.025	0.180 113 445	0.063 337 561	0.051 517 501	0.043 989 239	0.038 058 536
“Exact”	0.236 986 800	0.068 390 786	0.051 711 676	0.044 014 438	0.038 072 838

on $[2, 3]$ yields the results in [Table 3.1.9](#). Applying the Euler semilinear method with

$$y = ue^{-x^3} \quad \text{and} \quad u' = e^{x^3}(1 + u^2e^{-2x^3}), \quad u(2) = 2e^8$$

yields the results in [Table 3.1.10](#). Noting the close agreement among the three rows of [Table 3.1.9](#) (at least for larger values of x) and the lack of any such agreement among the rows of [Table 3.1.10](#), we conclude that Euler's method is better than the Euler semilinear method for [\(3.1.26\)](#). Comparing the results with the exact values supports this conclusion.

In the next two sections we'll study other numerical methods for solving initial value problems, called the *improved Euler method*, the *midpoint method*, *Heun's method* and the *Runge-Kutta method*. If the initial value problem is semilinear as in [\(3.1.19\)](#), we also have the option of using

Table 3.1.10: Numerical solution of $y' + 3x^2y = 1 + y^2$, $y(2) = 2$, by the Euler semilinear method.

h	x				
	2.2	2.4	2.6	2.8	3.0
0.1	0.214 501 852	0.032 487 396	0.017 332 058	0.011 819 555	0.008 065 020
0.05	0.222 599 468	0.047 079 261	0.030 750 181	0.023 720 670	0.018 472 302
0.025	0.228 942 240	0.056 825 805	0.040 189 920	0.032 679 767	0.026 931 099
“Exact”	0.236 986 800	0.068 390 786	0.051 711 676	0.044 014 438	0.038 072 838

variation of parameters and then applying the given numerical method to the initial value problem (3.1.21) for u . By analogy with the terminology used here, we'll call the resulting procedure *the improved Euler semilinear method*, the *midpoint semilinear method*, *Heun's semilinear method* or the *Runge-Kutta semilinear method*, as the case may be.

3.1 Exercises

You may want to save the results of these exercises, since we'll revisit in the next two sections. In Exercises 2–6 use Euler's method to find approximate values of the solution of the given initial value problem at the points $x_i = x_0 + ih$, where x_0 is the point where the initial condition is imposed and $i = 1, 2, 3$. The purpose of these exercises is to familiarize you with the computational procedure of Euler's method.

2. C $y' = 2x^2 + 3y^2 - 2, \quad y(2) = 1; \quad h = 0.05$
3. C $y' = y + \sqrt{x^2 + y^2}, \quad y(0) = 1; \quad h = 0.1$
4. C $y' + 3y = x^2 - 3xy + y^2, \quad y(0) = 2; \quad h = 0.05$
5. C $y' = \frac{1+x}{1-y^2}, \quad y(2) = 3; \quad h = 0.1$
6. C $y' + x^2y = \sin xy, \quad y(1) = \pi; \quad h = 0.2$
7. C Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 3y = 7e^{4x}, \quad y(0) = 2$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution $y = e^{4x} + e^{-3x}$, which can be obtained by the method of Section 2.1. Present your results in a table like Table 3.1.1.

8. C Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + \frac{2}{x}y = \frac{3}{x^3} + 1, \quad y(1) = 1$$

at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$. Compare these approximate values with the values of the exact solution

$$y = \frac{1}{3x^2}(9 \ln x + x^3 + 2),$$

which can be obtained by the method of Section 2.1. Present your results in a table like Table 3.1.1.

9. C Use Euler's method with step sizes $h = 0.05$, $h = 0.025$, and $h = 0.0125$ to find approximate values of the solution of the initial value problem

$$y' = \frac{y^2 + xy - x^2}{x^2}, \quad y(1) = 2$$

at $x = 1.0, 1.05, 1.10, 1.15, \dots, 1.5$. Compare these approximate values with the values of the exact solution

$$y = \frac{x(1 + x^2/3)}{1 - x^2/3}$$

obtained in Example 2.4.3. Present your results in a table like Table 3.1.1.

10. C In [Example 2.2.3](#) it was shown that

$$y^5 + y = x^2 + x - 4$$

is an implicit solution of the initial value problem

$$y' = \frac{2x+1}{5y^4+1}, \quad y(2) = 1. \quad (\text{A})$$

Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of (A) at $x = 2.0, 2.1, 2.2, 2.3, \dots, 3.0$. Present your results in tabular form. To check the error in these approximate values, construct another table of values of the residual

$$R(x, y) = y^5 + y - x^2 - x + 4$$

for each value of (x, y) appearing in the first table.

11. C You can see from [Example 2.5.1](#) that

$$x^4y^3 + x^2y^5 + 2xy = 4$$

is an implicit solution of the initial value problem

$$y' = -\frac{4x^3y^3 + 2xy^5 + 2y}{3x^4y^2 + 5x^2y^4 + 2x}, \quad y(1) = 1. \quad (\text{A})$$

Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of (A) at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$. Present your results in tabular form. To check the error in these approximate values, construct another table of values of the residual

$$R(x, y) = x^4y^3 + x^2y^5 + 2xy - 4$$

for each value of (x, y) appearing in the first table.

12. C Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$(3y^2 + 4y)y' + 2x + \cos x = 0, \quad y(0) = 1; \quad (\text{Exercise 2.2.17})$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$.

13. C Use Euler's method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + \frac{(y+1)(y-1)(y-2)}{x+1} = 0, \quad y(1) = 0 \text{ } \textit{text(Exercise 2.2.18)}$$

at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$.

14. C Use Euler's method and the Euler semilinear method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 3y = 7e^{-3x}, \quad y(0) = 6$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution $y = e^{-3x}(7x + 6)$, which can be obtained by the method of Section 2.1. Do you notice anything special about the results? Explain.

The linear initial value problems in Exercises 16–21 can't be solved exactly in terms of known elementary functions. In each exercise, use Euler's method and the Euler semilinear methods with the indicated step sizes to find approximate values of the solution of the given initial value problem at 11 equally spaced points (including the endpoints) in the interval.

16. C $y' - 2y = \frac{1}{1+x^2}, \quad y(2) = 2; \quad h = 0.1, 0.05, 0.025 \text{ on } [2, 3]$
17. C $y' + 2xy = x^2, \quad y(0) = 3 \text{ (Exercise 2.1.45); } \quad h = 0.2, 0.1, 0.05 \text{ on } [0, 2]$
18. C $y' + \frac{1}{x}y = \frac{\sin x}{x^2}, \quad y(1) = 2; \text{ (Exercise 2.1.46); } \quad h = 0.2, 0.1, 0.05 \text{ on } [1, 3]$
19. C $y' + y = \frac{e^{-x} \tan x}{x}, \quad y(1) = 0; \text{ (Exercise 2.1.47); } \quad h = 0.05, 0.025, 0.0125 \text{ on } [1, 1.5]$
20. C $y' + \frac{2x}{1+x^2}y = \frac{e^x}{(1+x^2)^2}, \quad y(0) = 1; \text{ (Exercise 2.1.48); } \quad h = 0.2, 0.1, 0.05 \text{ on } [0, 2]$
21. C $xy' + (x+1)y = e^{x^2}, \quad y(1) = 2; \text{ (Exercise 2.1.49); } \quad h = 0.05, 0.025, 0.0125 \text{ on } [1, 1.5]$

In Exercises 23–25, use Euler's method and the Euler semilinear method with the indicated step sizes to find approximate values of the solution of the given initial value problem at 11 equally spaced points (including the endpoints) in the interval.

23. C $y' + 3y = xy^2(y+1), \quad y(0) = 1; \quad h = 0.1, 0.05, 0.025 \text{ on } [0, 1]$
24. C $y' - 4y = \frac{x}{y^2(y+1)}, \quad y(0) = 1; \quad h = 0.1, 0.05, 0.025 \text{ on } [0, 1]$
25. C $y' + 2y = \frac{x^2}{1+y^2}, \quad y(2) = 1; \quad h = 0.1, 0.05, 0.025 \text{ on } [2, 3]$

26. **NUMERICAL QUADRATURE.** The fundamental theorem of calculus says that if f is continuous on a closed interval $[a, b]$ then it has an antiderivative F such that $F'(x) = f(x)$ on $[a, b]$ and

$$\int_a^b f(x) dx = F(b) - F(a). \quad (\text{A})$$

This solves the problem of evaluating a definite integral if the integrand f has an antiderivative that can be found and evaluated easily. However, if f doesn't have this property, (A) doesn't provide a useful way to evaluate the definite integral. In this case we must resort to approximate methods. There's a class of such methods called *numerical quadrature*, where the approximation takes the form

$$\int_a^b f(x) dx \approx \sum_{i=0}^n c_i f(x_i), \quad (\text{B})$$

where $a = x_0 < x_1 < \dots < x_n = b$ are suitably chosen points and $c_0, c_1, \dots, c_n$ are suitably chosen constants. We call (B) a *quadrature formula*.

- (a) Derive the quadrature formula

$$\int_a^b f(x) dx \approx h \sum_{i=0}^{n-1} f(a + ih) \quad (\text{where } h = (b - a)/n) \quad (\text{C})$$

by applying Euler's method to the initial value problem

$$y' = f(x), \quad y(a) = 0.$$

- (b) The quadrature formula (C) is sometimes called *the left rectangle rule*. Draw a figure that justifies this terminology.
- (c) **[L]** For several choices of a , b , and A , apply (C) to $f(x) = A$ with $n = 10, 20, 40, 80, 160, 320$. Compare your results with the exact answers and explain what you find.
- (d) **[L]** For several choices of a , b , A , and B , apply (C) to $f(x) = A + Bx$ with $n = 10, 20, 40, 80, 160, 320$. Compare your results with the exact answers and explain what you find.

3.2 The Improved Euler Method and Related Methods

In Section 3.1 we saw that the global truncation error of Euler's method is $O(h)$, which would seem to imply that we can achieve arbitrarily accurate results with Euler's method by simply choosing the step size sufficiently small. However, this isn't a good idea, for two reasons. First, after a certain point decreasing the step size will increase roundoff errors to the point where the accuracy will deteriorate rather than improve. The second and more important reason is that in most applications of numerical methods to an initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0, \quad (3.2.1)$$

the expensive part of the computation is the evaluation of f . Therefore we want methods that give good results for a given number of such evaluations. This is what motivates us to look for numerical methods better than Euler's.

Table 3.2.1: Approximations to $e \approx 2.718281828$ obtained by three numerical methods.

Evaluations of f	Euler	Improved Euler	Runge-Kutta
12	2.613035290	2.707188994	2.718069764
24	2.663731258	2.715327371	2.718266612
48	2.690496599	2.717519565	2.718280809

To clarify this point, suppose we want to approximate the value of e by applying Euler's method to the initial value problem

$$y' = y, \quad y(0) = 1, \quad (\text{with solution } y = e^x)$$

on $[0, 1]$, with $h = 1/12$, $1/24$, and $1/48$, respectively. Since each step in Euler's method requires one evaluation of f , the number of evaluations of f in each of these attempts is $n = 12$, 24 , and 48 , respectively. In each case we accept y_n as an approximation to e . The second column of Table 3.2.1 shows the results.

In this section we'll study the *improved Euler method*, which requires two evaluations of f at each step. We've used this method with $h = 1/6$, $1/12$, and $1/24$. The required number of evaluations of f were 12, 24, and 48, as in the three applications of Euler's method; however, you can see from the third column of Table 3.2.1 that the approximation to e obtained by the improved Euler method with only 12 evaluations of f is better than the approximation obtained by Euler's method with 48 evaluations.

In Section 3.3 we'll study the *Runge-Kutta method*, which requires four evaluations of f at each step. We've used this method with $h = 1/3$, $1/6$, and $1/12$. The required number of evaluations of f were again 12, 24, and 48, as in the three applications of Euler's method and the improved Euler method; however, you can see from the fourth column of Table 3.2.1 that the approximation to e obtained by the Runge-Kutta method with only 12 evaluations of f is better than the approximation obtained by the improved Euler method with 48 evaluations.

The Improved Euler Method

The *improved Euler method* for solving the initial value problem (3.2.1) is based on approximating the integral curve of (3.2.1) at $(x_i, y(x_i))$ by the line through $(x_i, y(x_i))$ with slope

$$m_i = \frac{f(x_i, y(x_i)) + f(x_{i+1}, y(x_{i+1}))}{2},$$

that is, m_i is the average of the slopes of the tangents to the integral curve at the endpoints of $[x_i, x_{i+1}]$. The equation of the approximating line is therefore

$$y = y(x_i) + \frac{f(x_i, y(x_i)) + f(x_{i+1}, y(x_{i+1}))}{2}(x - x_i). \quad (3.2.2)$$

Setting $x = x_{i+1} = x_i + h$ in (3.2.2) yields

$$y_{i+1} = y(x_i) + \frac{h}{2}(f(x_i, y(x_i)) + f(x_{i+1}, y(x_{i+1}))) \quad (3.2.3)$$

as an approximation to $y(x_{i+1})$. As in our derivation of Euler's method, we replace $y(x_i)$ (unknown if $i > 0$) by its approximate value y_i ; then (3.2.3) becomes

$$y_{i+1} = y_i + \frac{h}{2}(f(x_i, y_i) + f(x_{i+1}, y(x_{i+1}))).$$

However, this still won't work, because we don't know $y(x_{i+1})$, which appears on the right. We overcome this by replacing $y(x_{i+1})$ by $y_i + hf(x_i, y_i)$, the value that the Euler method would assign to y_{i+1} . Thus, the improved Euler method starts with the known value $y(x_0) = y_0$ and computes $y_1, y_2, \dots, y_n$ successively with the formula

$$y_{i+1} = y_i + \frac{h}{2} (f(x_i, y_i) + f(x_{i+1}, y_i + hf(x_i, y_i))). \quad (3.2.4)$$

The computation indicated here can be conveniently organized as follows: given y_i , compute

$$\begin{aligned} k_{1,i} &= f(x_i, y_i), \\ k_{2,i} &= f(x_i + h, y_i + hk_{1,i}), \\ y_{i+1} &= y_i + \frac{h}{2}(k_{1,i} + k_{2,i}). \end{aligned}$$

The improved Euler method requires two evaluations of $f(x, y)$ per step, while Euler's method requires only one. However, we'll see at the end of this section that if f satisfies appropriate assumptions, the local truncation error with the improved Euler method is $O(h^3)$, rather than $O(h^2)$ as with Euler's method. Therefore the global truncation error with the improved Euler method is $O(h^2)$; however, we won't prove this.

We note that the magnitude of the local truncation error in the improved Euler method and other methods discussed in this section is determined by the third derivative y''' of the solution of the initial value problem. Therefore the local truncation error will be larger where $|y'''|$ is large, or smaller where $|y'''|$ is small.

The next example, which deals with the initial value problem considered in [Example 3.1.1](#), illustrates the computational procedure indicated in the improved Euler method.

EXAMPLE 3.2.1. Use the improved Euler method with $h = 0.1$ to find approximate values of the solution of the initial value problem

$$y' + 2y = x^3 e^{-2x}, \quad y(0) = 1 \quad (3.2.5)$$

at $x = 0.1, 0.2, 0.3$.

SOLUTION As in [Example 3.1.1](#), we rewrite (3.2.5) as

$$y' = -2y + x^3 e^{-2x}, \quad y(0) = 1,$$

which is of the form (3.2.1), with

$$f(x, y) = -2y + x^3 e^{-2x}, \quad x_0 = 0, \text{ and } y_0 = 1.$$

The improved Euler method yields

$$\begin{aligned} k_{1,0} &= f(x_0, y_0) = f(0, 1) = -2, \\ k_{2,0} &= f(x_1, y_0 + hk_{1,0}) = f(.1, 1 + (.1)(-2)) \\ &= f(.1, .8) = -2(.8) + (.1)^3 e^{-.2} = -1.599181269, \\ y_1 &= y_0 + \frac{h}{2}(k_{1,0} + k_{2,0}) = 1 + (.05)(-2 - 1.599181269) = .820040937, \\ k_{1,1} &= f(x_1, y_1) = f(.1, .820040937) = -2(.820040937) + (.1)^3 e^{-.2} = -1.639263142, \end{aligned}$$

Table 3.2.2: Numerical solution of $y' + 2y = x^3e^{-2x}$, $y(0) = 1$, by Euler's method and the improved Euler method.

h	x				
	0.2	0.4	0.6	0.8	1.0
Euler					
0.1	0.640 081 873	0.411 563 195	0.270 299 502	0.186 654 593	0.139 778 910
0.05	0.656 266 437	0.432 887 056	0.291 404 256	0.205 105 754	0.154 715 925
Improved Euler					
0.1	0.672 734 445	0.455 160 637	0.313 970 920	0.225 267 702	0.171 388 070
0.05	0.671 086 455	0.452 890 616	0.311 652 239	0.223 194 281	0.169 680 673
Exact	0.670 588 174	0.452 204 669	0.310 952 904	0.222 570 721	0.169 169 104

$$\begin{aligned}
k_{2,1} &= f(x_2, y_1 + hk_{1,1}) = f(.2, .820040937 + .1(-1.639263142)) \\
&= f(.2, .656114622) = -2(.656114622) + (.2)^3e^{-.4} = -1.306866684, \\
y_2 &= y_1 + \frac{h}{2}(k_{1,1} + k_{2,1}) = .820040937 + (.05)(-1.639263142 - 1.306866684) = .672734445, \\
k_{1,2} &= f(x_2, y_2) = f(.2, .672734445) = -2(.672734445) + (.2)^3e^{-.4} = -1.340106330, \\
k_{2,2} &= f(x_3, y_2 + hk_{1,2}) = f(.3, .672734445 + .1(-1.340106330)), \\
&= f(.3, .538723812) = -2(.538723812) + (.3)^3e^{-.6} = -1.062629710, \\
y_3 &= y_2 + \frac{h}{2}(k_{1,2} + k_{2,2}) = .672734445 + (.05)(-1.340106330 - 1.062629710) = .552597643.
\end{aligned}$$

EXAMPLE 3.2.2. Table 3.2.2 shows results of using the improved Euler method with step sizes $h = 0.1$ and $h = 0.05$ to find approximate values of the solution of the initial value problem

$$y' + 2y = x^3e^{-2x}, \quad y(0) = 1$$

at $x = 0.2, 0.4, \dots, 1.0$. For comparison, it also shows the corresponding approximate values obtained with Euler's method in Example 3.1.2, and the values of the exact solution

$$y = \frac{e^{-2x}}{4}(x^4 + 4).$$

The results obtained by the improved Euler method with $h = 0.1$ are better than those obtained by Euler's method with $h = 0.05$.

EXAMPLE 3.2.3. Table 3.2.3 shows analogous results for the nonlinear initial value problem

$$y' = -2y^2 + xy + x^2, \quad y(0) = 1.$$

We applied Euler's method to this problem in Example 3.1.3.

Table 3.2.3: Numerical solution of $y' = -2y^2 + xy + x^2$, $y(0) = 1$, by Euler's method and the improved Euler method.

h	x				
	0.2	0.4	0.6	0.8	1.0
Euler					
0.1	0.681 000 000	0.559 628 676	0.529 820 120	0.569 732 776	0.675 192 037
0.05	0.707 795 377	0.587 454 526	0.557 143 535	0.596 951 988	0.701 764 495
Improved Euler					
0.1	0.733 430 846	0.615 961 841	0.586 006 935	0.626 008 824	0.730 069 610
0.05	0.730 556 677	0.612 884 493	0.582 927 224	0.622 898 279	0.726 985 837
"Exact"	0.729 641 890	0.611 901 791	0.581 942 225	0.621 907 458	0.726 015 790

EXAMPLE 3.2.4. Use step sizes $h = 0.2$, $h = 0.1$, and $h = 0.05$ to find approximate values of the solution of

$$y' - 2xy = 1, \quad y(0) = 3 \quad (3.2.6)$$

at $x = 0, 0.2, 0.4, 0.6, \dots, 2.0$ by **(a)** the improved Euler method; **(b)** the improved Euler semilinear method. (We used Euler's method and the Euler semilinear method on this problem in [Example 3.1.4](#).)

SOLUTION(a) Rewriting (3.2.6) as

$$y' = 1 + 2xy, \quad y(0) = 3$$

and applying the improved Euler method with $f(x, y) = 1 + 2xy$ yields the results shown in [Table 3.2.4](#).

SOLUTION(b) Since $y_1 = e^{x^2}$ is a solution of the complementary equation $y' - 2xy = 0$, we can apply the improved Euler semilinear method to (3.2.6), with

$$y = ue^{x^2} \quad \text{and} \quad u' = e^{-x^2}, \quad u(0) = 3.$$

The results listed in [Table 3.2.5](#) are clearly better than those obtained by the improved Euler method.

Table 3.2.4: Numerical solution of $y' - 2xy = 1$, $y(0) = 3$, by the improved Euler method.

h	x				
	0.4	0.8	1.2	1.6	2.0
0.2	3.964 659 200	6.900 088 156	15.708 954 42	46.958 915 75	184.563 776 3
0.1	3.966 340 117	6.928 648 973	15.970 033 26	49.096 125 52	203.151 922 7
0.05	3.966 216 690	6.934 862 367	16.041 904 86	49.754 131 06	209.464 744 5
“Exact”	3.966 059 348	6.936 700 945	16.067 111 68	50.000 377 78	211.954 462 2

Table 3.2.5: Numerical solution of $y' - 2xy = 1$, $y(0) = 3$, by the improved Euler semilinear method.

h	x				
	0.4	0.8	1.2	1.6	2.0
0.2	3.963 383 070	6.931 355 329	16.059 110 51	49.989 741 53	211.941 217 8
0.1	3.965 392 084	6.935 366 847	16.065 111 60	49.997 713 00	211.951 134 4
0.05	3.965 892 644	6.936 367 564	16.066 611 67	49.999 711 23	211.953 629 2
“Exact”	3.966 059 348	6.936 700 945	16.067 111 68	50.000 377 78	211.954 462 2

A Family of Methods with $O(h^3)$ Local Truncation Error

We'll now derive a class of methods with $O(h^3)$ local truncation error for solving (3.2.1). For simplicity, we assume that f , f_x , f_y , f_{xx} , f_{yy} , and f_{xy} are continuous and bounded for all (x, y) . This implies that if y is the solution of (3.2.1) then y'' and y''' are bounded (Exercise 34).

We begin by approximating the integral curve of (3.2.1) at $(x_i, y(x_i))$ by the line through $(x_i, y(x_i))$ with slope

$$m_i = \sigma y'(x_i) + \rho y'(x_i + \theta h),$$

where σ , ρ , and θ are constants that we'll soon specify; however, we insist at the outset that $0 < \theta \leq 1$, so that

$$x_i < x_i + \theta h \leq x_{i+1}.$$

The equation of the approximating line is

$$\begin{aligned} y &= y(x_i) + m_i(x - x_i) \\ &= y(x_i) + [\sigma y'(x_i) + \rho y'(x_i + \theta h)](x - x_i). \end{aligned} \quad (3.2.7)$$

Setting $x = x_{i+1} = x_i + h$ in (3.2.7) yields

$$\hat{y}_{i+1} = y(x_i) + h[\sigma y'(x_i) + \rho y'(x_i + \theta h)]$$

as an approximation to $y(x_{i+1})$.

To determine σ , ρ , and θ so that the error

$$\begin{aligned} E_i &= y(x_{i+1}) - \hat{y}_{i+1} \\ &= y(x_{i+1}) - y(x_i) - h[\sigma y'(x_i) + \rho y'(x_i + \theta h)] \end{aligned} \quad (3.2.8)$$

in this approximation is $O(h^3)$, we begin by recalling from Taylor's theorem that

$$y(x_{i+1}) = y(x_i) + hy'(x_i) + \frac{h^2}{2}y''(x_i) + \frac{h^3}{6}y'''(\hat{x}_i),$$

where $\hat{x}_i$ is in (x_i, x_{i+1}) . Since y''' is bounded this implies that

$$y(x_{i+1}) - y(x_i) - hy'(x_i) - \frac{h^2}{2}y''(x_i) = O(h^3).$$

Comparing this with (3.2.8) shows that $E_i = O(h^3)$ if

$$\sigma y'(x_i) + \rho y'(x_i + \theta h) = y'(x_i) + \frac{h}{2}y''(x_i) + O(h^2). \quad (3.2.9)$$

However, applying Taylor's theorem to y' shows that

$$y'(x_i + \theta h) = y'(x_i) + \theta h y''(x_i) + \frac{(\theta h)^2}{2}y'''(\bar{x}_i),$$

where $\bar{x}_i$ is in $(x_i, x_i + \theta h)$. Since y''' is bounded, this implies that

$$y'(x_i + \theta h) = y'(x_i) + \theta h y''(x_i) + O(h^2).$$

Substituting this into (3.2.9) and noting that the sum of two $O(h^2)$ terms is again $O(h^2)$ shows that $E_i = O(h^3)$ if

$$(\sigma + \rho)y'(x_i) + \rho \theta h y''(x_i) = y'(x_i) + \frac{h}{2}y''(x_i),$$

which is true if

$$\sigma + \rho = 1 \quad \text{and} \quad \rho \theta = \frac{1}{2}. \quad (3.2.10)$$

Since $y' = f(x, y)$, we can now conclude from (3.2.8) that

$$y(x_{i+1}) = y(x_i) + h[\sigma f(x_i, y_i) + \rho f(x_i + \theta h, y(x_i + \theta h))] + O(h^3) \quad (3.2.11)$$

if σ , ρ , and θ satisfy (3.2.10). However, this formula would not be useful even if we knew $y(x_i)$ exactly (as we would for $i = 0$), since we still wouldn't know $y(x_i + \theta h)$ exactly. To overcome this difficulty, we again use Taylor's theorem to write

$$y(x_i + \theta h) = y(x_i) + \theta h y'(x_i) + \frac{h^2}{2}y''(\tilde{x}_i),$$

where $\tilde{x}_i$ is in $(x_i, x_i + \theta h)$. Since $y'(x_i) = f(x_i, y(x_i))$ and y'' is bounded, this implies that

$$|y(x_i + \theta h) - y(x_i) - \theta h f(x_i, y(x_i))| \leq Kh^2 \quad (3.2.12)$$

for some constant K . Since f_y is bounded, the mean value theorem implies that

$$|f(x_i + \theta h, u) - f(x_i + \theta h, v)| \leq M|u - v|$$

for some constant M . Letting

$$u = y(x_i + \theta h) \quad \text{and} \quad v = y(x_i) + \theta h f(x_i, y(x_i))$$

and recalling (3.2.12) shows that

$$f(x_i + \theta h, y(x_i + \theta h)) = f(x_i + \theta h, y(x_i) + \theta h f(x_i, y(x_i))) + O(h^2).$$

Substituting this into (3.2.11) yields

$$y(x_{i+1}) = y(x_i) + h[\sigma f(x_i, y(x_i)) + \rho f(x_i + \theta h, y(x_i) + \theta h f(x_i, y(x_i)))] + O(h^3).$$

This implies that the formula

$$y_{i+1} = y_i + h[\sigma f(x_i, y_i) + \rho f(x_i + \theta h, y_i + \theta h f(x_i, y_i))]$$

has $O(h^3)$ local truncation error if σ , ρ , and θ satisfy (3.2.10). Substituting $\sigma = 1 - \rho$ and $\theta = 1/2\rho$ here yields

$$y_{i+1} = y_i + h \left[(1 - \rho)f(x_i, y_i) + \rho f \left(x_i + \frac{h}{2\rho}, y_i + \frac{h}{2\rho} f(x_i, y_i) \right) \right]. \quad (3.2.13)$$

The computation indicated here can be conveniently organized as follows: given y_i , compute

$$\begin{aligned} k_{1,i} &= f(x_i, y_i), \\ k_{2,i} &= f \left(x_i + \frac{h}{2\rho}, y_i + \frac{h}{2\rho} k_{1,i} \right), \\ y_{i+1} &= y_i + h[(1 - \rho)k_{1,i} + \rho k_{2,i}]. \end{aligned}$$

Consistent with our requirement that $0 < \theta < 1$, we require that $\rho \geq 1/2$. Letting $\rho = 1/2$ in (3.2.13) yields the improved Euler method (3.2.4). Letting $\rho = 3/4$ yields *Heun's method*,

$$y_{i+1} = y_i + h \left[\frac{1}{4}f(x_i, y_i) + \frac{3}{4}f \left(x_i + \frac{2}{3}h, y_i + \frac{2}{3}hf(x_i, y_i) \right) \right],$$

which can be organized as

$$\begin{aligned} k_{1,i} &= f(x_i, y_i), \\ k_{2,i} &= f \left(x_i + \frac{2h}{3}, y_i + \frac{2h}{3}k_{1,i} \right), \\ y_{i+1} &= y_i + \frac{h}{4}(k_{1,i} + 3k_{2,i}). \end{aligned}$$

Letting $\rho = 1$ yields the *midpoint method*,

$$y_{i+1} = y_i + hf \left(x_i + \frac{h}{2}, y_i + \frac{h}{2}f(x_i, y_i) \right),$$

which can be organized as

$$\begin{aligned} k_{1,i} &= f(x_i, y_i), \\ k_{2,i} &= f \left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{1,i} \right), \\ y_{i+1} &= y_i + hk_{2,i}. \end{aligned}$$

Examples involving the midpoint method and Heun's method are given in Exercises 26–33.

3.2 Exercises

Most of the following numerical exercises involve initial value problems considered in the exercises in Section 3.1. You'll find it instructive to compare the results that you obtain here with the corresponding results that you obtained in Section 3.1.

In Exercises 2–6 use the improved Euler method to find approximate values of the solution of the given initial value problem at the points $x_i = x_0 + ih$, where x_0 is the point where the initial condition is imposed and $i = 1, 2, 3$.

2. C $y' = 2x^2 + 3y^2 - 2, \quad y(2) = 1; \quad h = 0.05$
3. C $y' = y + \sqrt{x^2 + y^2}, \quad y(0) = 1; \quad h = 0.1$
4. C $y' + 3y = x^2 - 3xy + y^2, \quad y(0) = 2; \quad h = 0.05$
5. C $y' = \frac{1+x}{1-y^2}, \quad y(2) = 3; \quad h = 0.1$
6. C $y' + x^2y = \sin xy, \quad y(1) = \pi; \quad h = 0.2$
7. C Use the improved Euler method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 3y = 7e^{4x}, \quad y(0) = 2$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution $y = e^{4x} + e^{-3x}$, which can be obtained by the method of Section 2.1. Present your results in a table like [Table 3.2.2](#).

8. C Use the improved Euler method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + \frac{2}{x}y = \frac{3}{x^3} + 1, \quad y(1) = 1$$

at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$. Compare these approximate values with the values of the exact solution

$$y = \frac{1}{3x^2}(9 \ln x + x^3 + 2)$$

which can be obtained by the method of Section 2.1. Present your results in a table like [Table 3.2.2](#).

9. C Use the improved Euler method with step sizes $h = 0.05$, $h = 0.025$, and $h = 0.0125$ to find approximate values of the solution of the initial value problem

$$y' = \frac{y^2 + xy - x^2}{x^2}, \quad y(1) = 2,$$

at $x = 1.0, 1.05, 1.10, 1.15, \dots, 1.5$. Compare these approximate values with the values of the exact solution

$$y = \frac{x(1 + x^2/3)}{1 - x^2/3}$$

obtained in [Example 2.4.3](#). Present your results in a table like [Table 3.2.2](#).

10. C In [Example 3.2.2](#) it was shown that

$$y^5 + y = x^2 + x - 4$$

is an implicit solution of the initial value problem

$$y' = \frac{2x+1}{5y^4+1}, \quad y(2) = 1. \quad (\text{A})$$

Use the improved Euler method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of (A) at $x = 2.0, 2.1, 2.2, 2.3, \dots, 3.0$. Present your results in tabular form. To check the error in these approximate values, construct another table of values of the residual

$$R(x, y) = y^5 + y - x^2 - x + 4$$

for each value of (x, y) appearing in the first table.

11. C You can see from [Example 2.5.1](#) that

$$x^4y^3 + x^2y^5 + 2xy = 4$$

is an implicit solution of the initial value problem

$$y' = -\frac{4x^3y^3 + 2xy^5 + 2y}{3x^4y^2 + 5x^2y^4 + 2x}, \quad y(1) = 1. \quad (\text{A})$$

Use the improved Euler method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of (A) at $x = 1.0, 1.14, 1.2, 1.3, \dots, 2.0$. Present your results in tabular form. To check the error in these approximate values, construct another table of values of the residual

$$R(x, y) = x^4y^3 + x^2y^5 + 2xy - 4$$

for each value of (x, y) appearing in the first table.

12. C Use the improved Euler method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$(3y^2 + 4y)y' + 2x + \cos x = 0, \quad y(0) = 1 \quad (\text{Exercise 2.2.17})$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$.

13. C Use the improved Euler method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + \frac{(y+1)(y-1)(y-2)}{x+1} = 0, \quad y(1) = 0 \quad (\text{Exercise 2.2.18})$$

at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$.

14. C Use the improved Euler method and the improved Euler semilinear method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 3y = e^{-3x}(1 - 2x), \quad y(0) = 2,$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution $y = e^{-3x}(2 + x - x^2)$, which can be obtained by the method of Section 2.1. Do you notice anything special about the results? Explain.

The linear initial value problems in Exercises 16–21 can't be solved exactly in terms of known elementary functions. In each exercise use the improved Euler and improved Euler semilinear methods with the indicated step sizes to find approximate values of the solution of the given initial value problem at 11 equally spaced points (including the endpoints) in the interval.

16. C $y' - 2y = \frac{1}{1+x^2}$, $y(2) = 2$; $h = 0.1, 0.05, 0.025$ on $[2, 3]$
17. C $y' + 2xy = x^2$, $y(0) = 3$; $h = 0.2, 0.1, 0.05$ on $[0, 2]$ (Exercise 2.1.45)
18. C $y' + \frac{1}{x}y = \frac{\sin x}{x^2}$, $y(1) = 2$, $h = 0.2, 0.1, 0.05$ on $[1, 3]$ (Exercise 2.1.46)
19. C $y' + y = \frac{e^{-x} \tan x}{x}$, $y(1) = 0$; $h = 0.05, 0.025, 0.0125$ on $[1, 1.5]$ (Exercise 2.1.47)
20. C $y' + \frac{2x}{1+x^2}y = \frac{e^x}{(1+x^2)^2}$, $y(0) = 1$; $h = 0.2, 0.1, 0.05$ on $[0, 2]$ (Exercise 2.1.48)
21. C $xy' + (x+1)y = e^{x^2}$, $y(1) = 2$; $h = 0.05, 0.025, 0.0125$ on $[1, 1.5]$ (Exercise 2.1.49)

In Exercises 23–25 use the improved Euler method and the improved Euler semilinear method with the indicated step sizes to find approximate values of the solution of the given initial value problem at 11 equally spaced points (including the endpoints) in the interval.

23. C $y' + 3y = xy^2(y+1)$, $y(0) = 1$; $h = 0.1, 0.05, 0.025$ on $[0, 1]$
24. C $y' - 4y = \frac{x}{y^2(y+1)}$, $y(0) = 1$; $h = 0.1, 0.05, 0.025$ on $[0, 1]$
25. C $y' + 2y = \frac{x^2}{1+y^2}$, $y(2) = 1$; $h = 0.1, 0.05, 0.025$ on $[2, 3]$
26. C Do Exercise 8 with “improved Euler method” replaced by “midpoint method.”
27. C Do Exercise 8 with “improved Euler method” replaced by “Heun’s method.”
28. C Do Exercise 9 with “improved Euler method” replaced by “midpoint method.”
29. C Do Exercise 9 with “improved Euler method” replaced by “Heun’s method.”
30. C Do Exercise 12 with “improved Euler method” replaced by “midpoint method.”
31. C Do Exercise 12 with “improved Euler method” replaced by “Heun’s method.”
32. C Do Exercise 13 with “improved Euler method” replaced by “midpoint method.”
33. C Do Exercise 13 with “improved Euler method” replaced by “Heun’s method.”
34. Show that if f , f_x , f_y , f_{xx} , f_{yy} , and f_{xy} are continuous and bounded for all (x, y) and y is the solution of the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0,$$

then y'' and y''' are bounded.

35. NUMERICAL QUADRATURE (see Exercise 3.1.26).

(a) Derive the quadrature formula

$$\int_a^b f(x) dx \approx .5h(f(a) + f(b)) + h \sum_{i=1}^{n-1} f(a + ih) \quad (\text{where } h = (b - a)/n) \quad (\text{A})$$

by applying the improved Euler method to the initial value problem

$$y' = f(x), \quad y(a) = 0.$$

- (b) The quadrature formula (A) is called *the trapezoid rule*. Draw a figure that justifies this terminology.
- (c) **[L]** For several choices of a , b , A , and B , apply (A) to $f(x) = A + Bx$, with $n = 10, 20, 40, 80, 160, 320$. Compare your results with the exact answers and explain what you find.
- (d) **[L]** For several choices of a , b , A , B , and C , apply (A) to $f(x) = A + Bx + Cx^2$, with $n = 10, 20, 40, 80, 160, 320$. Compare your results with the exact answers and explain what you find.

3.3 The Runge-Kutta Method

In general, if k is any positive integer and f satisfies appropriate assumptions, there are numerical methods with local truncation error $O(h^{k+1})$ for solving an initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0. \quad (3.3.1)$$

Moreover, it can be shown that a method with local truncation error $O(h^{k+1})$ has global truncation error $O(h^k)$. In Sections 3.1 and 3.2 we studied numerical methods where $k = 1$ and $k = 2$. We'll skip methods for which $k = 3$ and proceed to the *Runge-Kutta* method, the most widely used method, for which $k = 4$. The magnitude of the local truncation error is determined by the fifth derivative $y^{(5)}$ of the solution of the initial value problem. Therefore the local truncation error will be larger where $|y^{(5)}|$ is large, or smaller where $|y^{(5)}|$ is small. The Runge-Kutta method computes approximate values $y_1, y_2, \dots, y_n$ of the solution of (3.3.1) at $x_0, x_0 + h, \dots, x_0 + nh$ as follows: Given y_i , compute

$$\begin{aligned} k_{1,i} &= f(x_i, y_i), \\ k_{2,i} &= f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{1,i}\right), \\ k_{3,i} &= f\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}k_{2,i}\right), \\ k_{4,i} &= f(x_i + h, y_i + hk_{3,i}), \quad \text{and} \\ y_{i+1} &= y_i + \frac{h}{6}(k_{1,i} + 2k_{2,i} + 2k_{3,i} + k_{4,i}). \end{aligned}$$

The next example, which deals with the initial value problem considered in Examples 3.1.1 and 3.2.1, illustrates the computational procedure indicated in the Runge-Kutta method.

EXAMPLE 3.3.1. Use the Runge-Kutta method with $h = 0.1$ to find approximate values for the solution of the initial value problem

$$y' + 2y = x^3 e^{-2x}, \quad y(0) = 1, \quad (3.3.2)$$

at $x = 0.1, 0.2$.

SOLUTION Again we rewrite (3.3.2) as

$$y' = -2y + x^3 e^{-2x}, \quad y(0) = 1,$$

which is of the form (3.3.1), with

$$f(x, y) = -2y + x^3 e^{-2x}, \quad x_0 = 0, \text{ and } y_0 = 1.$$

The Runge-Kutta method yields

$$\begin{aligned} k_{1,0} &= f(x_0, y_0) = f(0, 1) = -2, \\ k_{2,0} &= f(x_0 + h/2, y_0 + hk_{1,0}/2) = f(.05, 1 + (.05)(-2)) \\ &= f(.05, .9) = -2(.9) + (.05)^3 e^{-.1} = -1.799886895, \\ k_{3,0} &= f(x_0 + h/2, y_0 + hk_{2,0}/2) = f(.05, 1 + (.05)(-1.799886895)) \\ &= f(.05, .910005655) = -2(.910005655) + (.05)^3 e^{-.1} = -1.819898206, \\ k_{4,0} &= f(x_0 + h, y_0 + hk_{3,0}) = f(.1, 1 + (.1)(-1.819898206)) \\ &= f(.1, .818010179) = -2(.818010179) + (.1)^3 e^{-.2} = -1.635201628, \\ y_1 &= y_0 + \frac{h}{6}(k_{1,0} + 2k_{2,0} + 2k_{3,0} + k_{4,0}), \\ &= 1 + \frac{.1}{6}(-2 + 2(-1.799886895) + 2(-1.819898206) - 1.635201628) = .818753803, \\ k_{1,1} &= f(x_1, y_1) = f(.1, .818753803) = -2(.818753803) + (.1)^3 e^{-.2} = -1.636688875, \\ k_{2,1} &= f(x_1 + h/2, y_1 + hk_{1,1}/2) = f(.15, .818753803 + (.05)(-1.636688875)) \\ &= f(.15, .736919359) = -2(.736919359) + (.15)^3 e^{-.3} = -1.471338457, \\ k_{3,1} &= f(x_1 + h/2, y_1 + hk_{2,1}/2) = f(.15, .818753803 + (.05)(-1.471338457)) \\ &= f(.15, .745186880) = -2(.745186880) + (.15)^3 e^{-.3} = -1.487873498, \\ k_{4,1} &= f(x_1 + h, y_1 + hk_{3,1}) = f(.2, .818753803 + (.1)(-1.487873498)) \\ &= f(.2, .669966453) = -2(.669966453) + (.2)^3 e^{-.4} = -1.334570346, \\ y_2 &= y_1 + \frac{h}{6}(k_{1,1} + 2k_{2,1} + 2k_{3,1} + k_{4,1}), \\ &= .818753803 + \frac{.1}{6}(-1.636688875 + 2(-1.471338457) + 2(-1.487873498) - 1.334570346) \\ &= .670592417. \end{aligned}$$

The Runge-Kutta method is sufficiently accurate for most applications.

EXAMPLE 3.3.2. Table 3.3.1 shows results of using the Runge-Kutta method with step sizes $h = 0.1$ and $h = 0.05$ to find approximate values of the solution of the initial value problem

$$y' + 2y = x^3 e^{-2x}, \quad y(0) = 1$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. For comparison, it also shows the corresponding approximate values obtained with the improved Euler method in Example 3.2.2, and the values of the exact solution

$$y = \frac{e^{-2x}}{4}(x^4 + 4).$$

Table 3.3.1: Numerical solution of $y' + 2y = x^3 e^{-2x}$, $y(0) = 1$, by the Runge-Kutta method and the improved Euler method.

h	x				
	0.2	0.4	0.6	0.8	1.0
Improved Euler					
0.1	0.672 734 445	0.455 160 637	0.313 970 920	0.225 267 702	0.171 388 070
0.05	0.671 086 455	0.452 890 616	0.311 652 239	0.223 194 281	0.169 680 673
Runge-Kutta					
0.1	0.670 592 417	0.452 210 430	0.310 958 768	0.222 575 989	0.169 173 489
0.05	0.670 588 418	0.452 205 001	0.310 953 242	0.222 571 024	0.169 169 356
Exact	0.670 588 174	0.452 204 669	0.310 952 904	0.222 570 721	0.169 169 104

Table 3.3.2: Numerical solution of $y' = -2y^2 + xy + x^2$, $y(0) = 1$, by the Runge-Kutta method and the improved Euler method.

h	x				
	0.2	0.4	0.6	0.8	1.0
Improved Euler					
0.1	0.733 430 846	0.615 961 841	0.586 006 935	0.626 008 824	0.730 069 610
0.05	0.730 556 677	0.612 884 493	0.582 927 224	0.622 898 279	0.726 985 837
Runge-Kutta					
0.1	0.729 644 487	0.611 903 380	0.581 943 210	0.621 908 378	0.726 017 378
0.05	0.729 642 155	0.611 901 969	0.581 942 342	0.621 907 553	0.726 015 908
“Exact”	0.729 641 890	0.611 901 791	0.581 942 225	0.621 907 458	0.726 015 790

The results obtained by the Runge-Kutta method are clearly better than those obtained by the improved Euler method in fact; the results obtained by the Runge-Kutta method with $h = 0.1$ are better than those obtained by the improved Euler method with $h = 0.05$.

EXAMPLE 3.3.3. Table 3.3.2 shows analogous results for the nonlinear initial value problem

$$y' = -2y^2 + xy + x^2, \quad y(0) = 1.$$

We applied the improved Euler method to this problem in Exercise 4.

EXAMPLE 3.3.4. Tables 3.3.3 and 3.3.4 show results obtained by applying the Runge-Kutta and Runge-Kutta semilinear methods to the initial value problem

$$y' - 2xy = 1, \quad y(0) = 3,$$

which we considered in Examples 3.1.4 and 3.2.4.

Table 3.3.3: Numerical solution of $y' - 2xy = 1$, $y(0) = 3$, by the Runge-Kutta method.

h	x				
	0.4	0.8	1.2	1.6	2.0
0.2	3.966 044 973	6.936 534 178	16.064 344 81	49.960 553 66	211.393 800 2
0.1	3.966 058 535	6.936 690 679	16.066 915 58	49.997 313 97	211.908 445 3
0.05	3.966 059 300	6.936 700 320	16.067 098 70	50.000 165 74	211.951 167 6
“Exact”	3.966 059 348	6.936 700 945	16.067 111 68	50.000 377 78	211.954 462 2

Table 3.3.4: Numerical solution of $y' - 2xy = 1$, $y(0) = 3$, by the Runge-Kutta semilinear method.

h	x				
	0.4	0.8	1.2	1.6	2.0
0.2	3.966 061 755	6.936 704 019	16.067 111 96	50.000 370 15	211.954 440 0
0.1	3.966 059 497	6.936 701 137	16.067 111 70	50.000 377 30	211.954 460 8
0.05	3.966 059 357	6.936 700 957	16.067 111 68	50.000 377 75	211.954 462 1
“Exact”	3.966 059 348	6.936 700 945	16.067 111 68	50.000 377 78	211.954 462 2

The Case Where x_0 Isn't The Left Endpoint

So far in this chapter we've considered numerical methods for solving an initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0 \quad (3.3.3)$$

on an interval $[x_0, b]$, for which x_0 is the left endpoint. We haven't discussed numerical methods for solving (3.3.3) on an interval $[a, x_0]$, for which x_0 is the right endpoint. To be specific, how can we obtain approximate values $y_{-1}, y_{-2}, \dots, y_{-n}$ of the solution of (3.3.3) at $x_0 - h, \dots, x_0 - nh$, where $h = (x_0 - a)/n$? Here's the answer to this question:

Consider the initial value problem

$$z' = -f(-x, z), \quad z(-x_0) = y_0, \quad (3.3.4)$$

on the interval $[-x_0, -a]$, for which $-x_0$ is the left endpoint. Use a numerical method to obtain approximate values $z_1, z_2, \dots, z_n$ of the solution of (3.3.4) at $-x_0 + h, -x_0 + 2h, \dots, -x_0 + nh = -a$. Then $y_{-1} = z_1, y_{-2} = z_2, \dots, y_{-n} = z_n$ are approximate values of the solution of (3.3.3) at $x_0 - h, x_0 - 2h, \dots, x_0 - nh = a$.

The justification for this answer is sketched in [Exercise 26](#). Note how easy it is to make the change the given problem (3.3.3) to the modified problem (3.3.4): first replace f by $-f$ and then replace x, x_0 , and y by $-x, -x_0$, and z , respectively.

EXAMPLE 3.3.5. Use the Runge-Kutta method with step size $h = 0.1$ to find approximate values of the solution of

$$(y - 1)^2 y' = 2x + 3, \quad y(1) = 4 \quad (3.3.5)$$

at $x = 0, 0.1, 0.2, \dots, 1$.

SOLUTION We first rewrite (3.3.5) in the form (3.3.3) as

$$y' = \frac{2x+3}{(y-1)^2}, \quad y(1) = 4. \quad (3.3.6)$$

Since the initial condition $y(1) = 4$ is imposed at the right endpoint of the interval $[0, 1]$, we apply the Runge-Kutta method to the initial value problem

$$z' = \frac{2x-3}{(z-1)^2}, \quad z(-1) = 4 \quad (3.3.7)$$

on the interval $[-1, 0]$. (You should verify that (3.3.7) is related to (3.3.6) as (3.3.4) is related to (3.3.3).) Table 3.3.5 shows the results. Changing the signs of the values of x yields the two rows of Table 3.3.6 (note that the x values are now decreasing). The exact values of y are given by

$$y = 1 + (3x^2 + 9x + 15)^{1/3}.$$

(Since the differential equation in (3.3.6) is separable, this formula can be obtained by the method of Section 2.2.)

Table 3.3.5: Numerical solution of $z' = \frac{2x-3}{(z-1)^2}$, $z(-1) = 4$, on $[-1, 0]$.

x	-0.8	-0.6	-0.4	-0.2	0.0
z	3.889 298 649	3.779 786 399	3.672 141 529	3.567 241 862	3.466 212 070

Table 3.3.6: Numerical solution of $(y-1)^2 y' = 2x+3$, $y(1) = 4$, on $[0, 1]$.

x	0.8	0.6	0.4	0.2	0.0
y	3.889 298 649	3.779 786 399	3.672 141 529	3.567 241 862	3.466 212 070
Exact	3.889 298 649	3.779 786 399	3.672 141 530	3.567 241 864	3.466 212 074

We leave it to you to develop a procedure for handling the numerical solution of (3.3.3) on an interval $[a, b]$ such that $a < x_0 < b$ (Exercises 29 and 30).

3.3 Exercises

Most of the following numerical exercises involve initial value problems considered in the exercises in Section 3.2. You'll find it instructive to compare the results that you obtain here with the corresponding results that you obtained in those sections.

In Exercises 2–6 use the Runge-Kutta method to find approximate values of the solution of the given initial value problem at the points $x_i = x_0 + ih$, where x_0 is the point where the initial condition is imposed and $i = 1, 2$.

2. C $y' = 2x^2 + 3y^2 - 2, \quad y(2) = 1; \quad h = 0.05$
3. C $y' = y + \sqrt{x^2 + y^2}, \quad y(0) = 1; \quad h = 0.1$
4. C $y' + 3y = x^2 - 3xy + y^2, \quad y(0) = 2; \quad h = 0.05$
5. C $y' = \frac{1+x}{1-y^2}, \quad y(2) = 3; \quad h = 0.1$
6. C $y' + x^2 y = \sin xy, \quad y(1) = \pi; \quad h = 0.2$

7. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 3y = 7e^{4x}, \quad y(0) = 2,$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution $y = e^{4x} + e^{-3x}$, which can be obtained by the method of Section 2.1. Present your results in a table like [Table 3.3.1](#).

8. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + \frac{2}{x}y = \frac{3}{x^3} + 1, \quad y(1) = 1$$

at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$. Compare these approximate values with the values of the exact solution

$$y = \frac{1}{3x^2}(9 \ln x + x^3 + 2),$$

which can be obtained by the method of Section 2.1. Present your results in a table like [Table 3.3.1](#).

9. C Use the Runge-Kutta method with step sizes $h = 0.05$, $h = 0.025$, and $h = 0.0125$ to find approximate values of the solution of the initial value problem

$$y' = \frac{y^2 + xy - x^2}{x^2}, \quad y(1) = 2$$

at $x = 1.0, 1.05, 1.10, 1.15, \dots, 1.5$. Compare these approximate values with the values of the exact solution

$$y = \frac{x(1 + x^2/3)}{1 - x^2/3},$$

which was obtained in [Example 2.2.3](#). Present your results in a table like [Table 3.3.1](#).

10. C In [Example 2.2.3](#) it was shown that

$$y^5 + y = x^2 + x - 4$$

is an implicit solution of the initial value problem

$$y' = \frac{2x + 1}{5y^4 + 1}, \quad y(2) = 1. \tag{A}$$

Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of (A) at $x = 2.0, 2.1, 2.2, 2.3, \dots, 3.0$. Present your results in tabular form. To check the error in these approximate values, construct another table of values of the residual

$$R(x, y) = y^5 + y - x^2 - x + 4$$

for each value of (x, y) appearing in the first table.

11. C You can see from [Example 2.5.1](#) that

$$x^4y^3 + x^2y^5 + 2xy = 4$$

is an implicit solution of the initial value problem

$$y' = -\frac{4x^3y^3 + 2xy^5 + 2y}{3x^4y^2 + 5x^2y^4 + 2x}, \quad y(1) = 1. \quad (\text{A})$$

Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of (A) at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$. Present your results in tabular form. To check the error in these approximate values, construct another table of values of the residual

$$R(x, y) = x^4y^3 + x^2y^5 + 2xy - 4$$

for each value of (x, y) appearing in the first table.

12. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$(3y^2 + 4y)y' + 2x + \cos x = 0, \quad y(0) = 1 \quad (\text{Exercise 2.2.17}),$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$.

13. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + \frac{(y+1)(y-1)(y-2)}{x+1} = 0, \quad y(1) = 0 \quad (\text{Exercise 2.2.18}),$$

at $x = 1.0, 1.1, 1.2, 1.3, \dots, 2.0$.

14. C Use the Runge-Kutta method and the Runge-Kutta semilinear method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of the initial value problem

$$y' + 3y = e^{-3x}(1 - 4x + 3x^2 - 4x^3), \quad y(0) = -3$$

at $x = 0, 0.1, 0.2, 0.3, \dots, 1.0$. Compare these approximate values with the values of the exact solution $y = -e^{-3x}(3 - x + 2x^2 - x^3 + x^4)$, which can be obtained by the method of Section 2.1. Do you notice anything special about the results? Explain.

The linear initial value problems in Exercises 16–21 can't be solved exactly in terms of known elementary functions. In each exercise use the Runge-Kutta and the Runge-Kutta semilinear methods with the indicated step sizes to find approximate values of the solution of the given initial value problem at 11 equally spaced points (including the endpoints) in the interval.

16. C $y' - 2y = \frac{1}{1+x^2}$, $y(2) = 2$; $h = 0.1, 0.05, 0.025$ on $[2, 3]$
17. C $y' + 2xy = x^2$, $y(0) = 3$; $h = 0.2, 0.1, 0.05$ on $[0, 2]$ (Exercise 2.1.45)
18. C $y' + \frac{1}{x}y = \frac{\sin x}{x^2}$, $y(1) = 2$; $h = 0.2, 0.1, 0.05$ on $[1, 3]$ (Exercise 2.1.46)
19. C $y' + y = \frac{e^{-x}\tan x}{x}$, $y(1) = 0$; $h = 0.05, 0.025, 0.0125$ on $[1, 1.5]$ (Exercise 2.1.47)

20. C $y' + \frac{2x}{1+x^2}y = \frac{e^x}{(1+x^2)^2}, \quad y(0) = 1; \quad h = 0.2, 0.1, 0.05 \text{ on } [0, 2]$ (Exercise 2.1.48)
21. C $xy' + (x+1)y = e^{x^2}, \quad y(1) = 2; \quad h = 0.05, 0.025, 0.0125 \text{ on } [1, 1.5]$ (Exercise 2.1.49)

In Exercises 23–25 use the Runge-Kutta method and the Runge-Kutta semilinear method with the indicated step sizes to find approximate values of the solution of the given initial value problem at 11 equally spaced points (including the endpoints) in the interval.

23. C $y' + 3y = xy^2(y+1), \quad y(0) = 1; \quad h = 0.1, 0.05, 0.025 \text{ on } [0, 1]$
24. C $y' - 4y = \frac{x}{y^2(y+1)}, \quad y(0) = 1; \quad h = 0.1, 0.05, 0.025 \text{ on } [0, 1]$
25. C $y' + 2y = \frac{x^2}{1+y^2}, \quad y(2) = 1; \quad h = 0.1, 0.05, 0.025 \text{ on } [2, 3]$
26. C Suppose $a < x_0$, so that $-x_0 < -a$. Use the chain rule to show that if z is a solution of

$$z' = -f(-x, z), \quad z(-x_0) = y_0,$$

on $[-x_0, -a]$, then $y = z(-x)$ is a solution of

$$y' = f(x, y), \quad y(x_0) = y_0,$$

on $[a, x_0]$.

27. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of

$$y' = \frac{y^2 + xy - x^2}{x^2}, \quad y(2) = -1$$

at $x = 1.1, 1.2, 1.3, \dots, 2.0$. Compare these approximate values with the values of the exact solution

$$y = \frac{x(4 - 3x^2)}{4 + 3x^2},$$

which can be obtained by referring to Example 2.4.3.

28. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of

$$y' = -x^2y - xy^2, \quad y(1) = 1$$

at $x = 0, 0.1, 0.2, \dots, 1$.

29. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of

$$y' + \frac{1}{x}y = \frac{7}{x^2} + 3, \quad y(1) = \frac{3}{2}$$

at $x = 0.5, 0.6, \dots, 1.5$. Compare these approximate values with the values of the exact solution

$$y = \frac{7 \ln x}{x} + \frac{3x}{2},$$

which can be obtained by the method discussed in Section 2.1.

30. C Use the Runge-Kutta method with step sizes $h = 0.1$, $h = 0.05$, and $h = 0.025$ to find approximate values of the solution of

$$xy' + 2y = 8x^2, \quad y(2) = 5$$

at $x = 1.0, 1.1, 1.2, \dots, 3.0$. Compare these approximate values with the values of the exact solution

$$y = 2x^2 - \frac{12}{x^2},$$

which can be obtained by the method discussed in Section 2.1.

31. NUMERICAL QUADRATURE (see Exercise 3.1.26).

(a) Derive the quadrature formula

$$\int_a^b f(x) dx \approx \frac{h}{6}(f(a) + f(b)) + \frac{h}{3} \sum_{i=1}^{n-1} f(a + ih) + \frac{2h}{3} \sum_{i=1}^n f(a + (2i-1)h/2) \quad (\text{A})$$

(where $h = (b - a)/n$) by applying the Runge-Kutta method to the initial value problem

$$y' = f(x), \quad y(a) = 0.$$

This quadrature formula is called *Simpson's Rule*.

- (b) L For several choices of a, b, A, B, C , and D apply (A) to $f(x) = A + Bx + Cx + Dx^3$, with $n = 10, 20, 40, 80, 160, 320$. Compare your results with the exact answers and explain what you find.
- (c) L For several choices of a, b, A, B, C, D , and E apply (A) to $f(x) = A + Bx + Cx^2 + Dx^3 + Ex^4$, with $n = 10, 20, 40, 80, 160, 320$. Compare your results with the exact answers and explain what you find.